

Readme Contents

1. What is this?
2. Who is this for?
3. Installation Guide
4. User Guide
5. Credits & License

1. What is this?

The Bcalc Firearm Management App is an open-source firearm, optics, and ammunition tracker with no cloud-based services or subscriptions.

You can enter multiple configurations of firearms, optics, and ammunition. For each configuration, you can track range session performance and conditions. Data reports such as rounds per firearm and top configurations by caliber are available.

An internet connection is not required once installed. No user data is transmitted.

2. Who is this for?

Hunters and hobbyists who prefer to use open-source applications. It's easy to track your firearm performance with different ammunition, whether purchased or hand-load.

3. Basic Linux Installation Guide

1. Linux

- download "bcalctracker" from the <https://github.com/briancalc/bcalctracker/releases> page
- to run using file manager, double click bcalctracker
- to run from the terminal `./bcalctracker`

2. Windows

- download "bcalctracker" from the <https://github.com/briancalc/bcalctracker/releases> page
- double click bcalctracker.exe

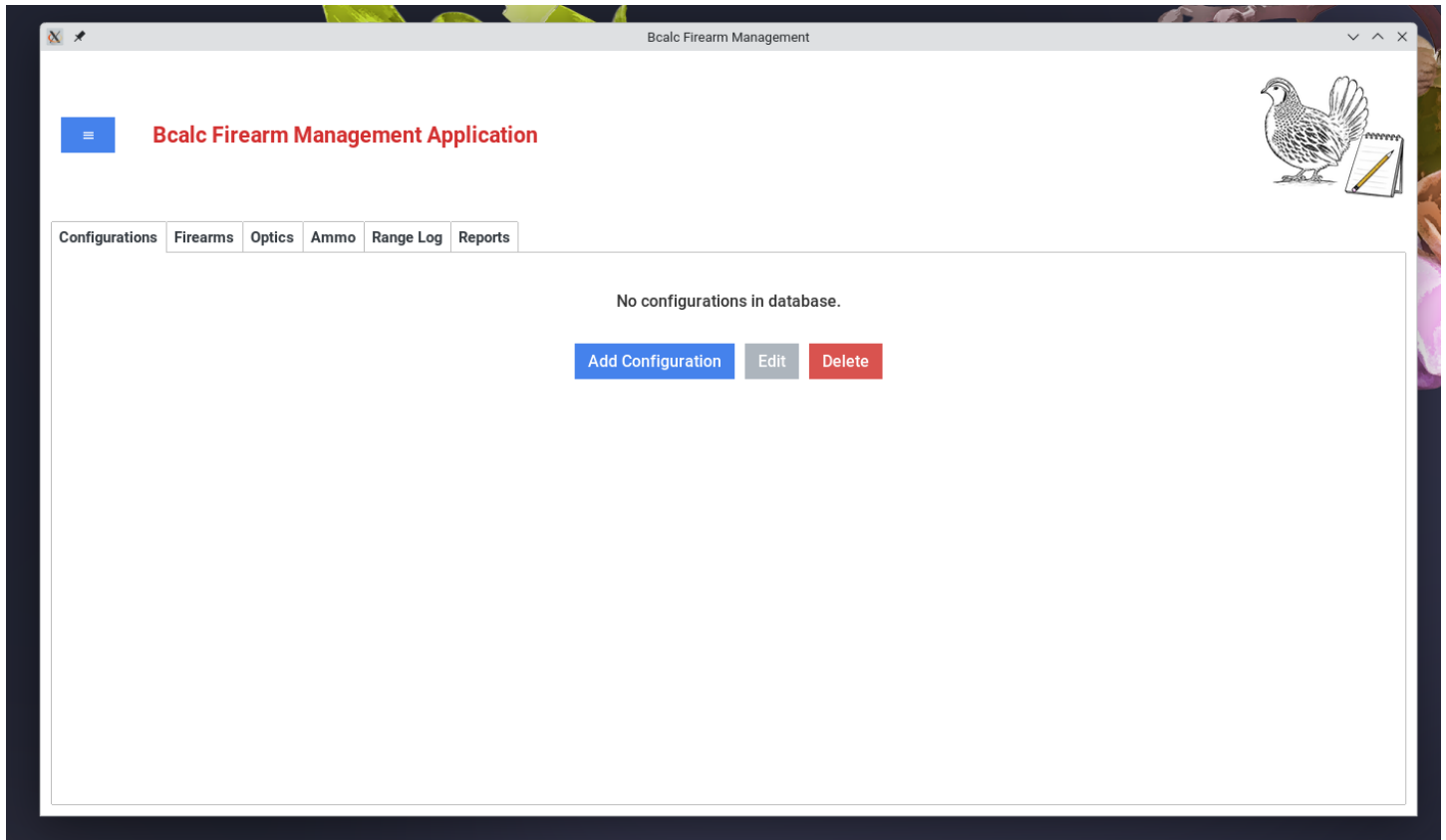
3. Mac

- in progress

4. User Guide

Step 1: Firearms

Configurations are what facilitate the collection of range session data. Configurations can consist of a firearm, an optic, and a specific ammunition or, at the minimum, just a firearm.



So, you need to enter a firearm: click on the Firearms tab and select the blue Add Firearm button. Firearm Name, Caliber Primary, and Barrel Length are required fields. Enter as many of the other fields as you would like, including attaching a photo. You can always return and edit the information later.

The screenshot displays the 'Bcalc Firearm Management' application window. A modal form titled 'Add New Firearm' is open, allowing users to input details for a new firearm. The form is organized into several sections: 'Basic Information' with fields for Firearm Name, Manufacturer, Type, Serial #, Caliber Primary, Caliber Secondary, and Barrel Length; 'Purchase Details (optional)' with fields for Purchase Date, Purchase Location, and Purchase Cost; 'Sold Status' with a checkbox for 'Mark as Sold'; and 'Photo & Notes' with a Firearm Photo upload field and a Notes field. The 'Firearm Name' field is pre-filled with 'BAR Mk3 Stalker', 'Manufacturer' with 'Browning', 'Type' with 'Rifle: Semi', and 'Serial #' with 'BAR Mk3 Stalker'. The 'Caliber Primary' field is set to '243 Win'. The 'Barrel Length (in)' field is empty. The 'Purchase Date' field is pre-filled with 'YYYY-MM-DD'. The 'Purchase Location' field is empty. The 'Purchase Cost' field is empty. The 'Mark as Sold' checkbox is unchecked. The 'Firearm Photo' field is empty, and the 'Notes' field is empty. The 'Browse...' button is visible next to the 'Firearm Photo' field. At the bottom of the modal, there are 'Save' and 'Cancel' buttons. Below the modal, in the background application window, there are 'Add Firearm', 'Edit', and 'Delete' buttons.

Add New Firearm

Basic Information

Firearm Name: Manufacturer:

Type: Serial #:

Caliber Primary: Caliber Secondary:

Barrel Length (in):

Purchase Details (optional)

Purchase Date: Purchase Location:

Purchase Cost:

Sold Status

☐ Mark as Sold

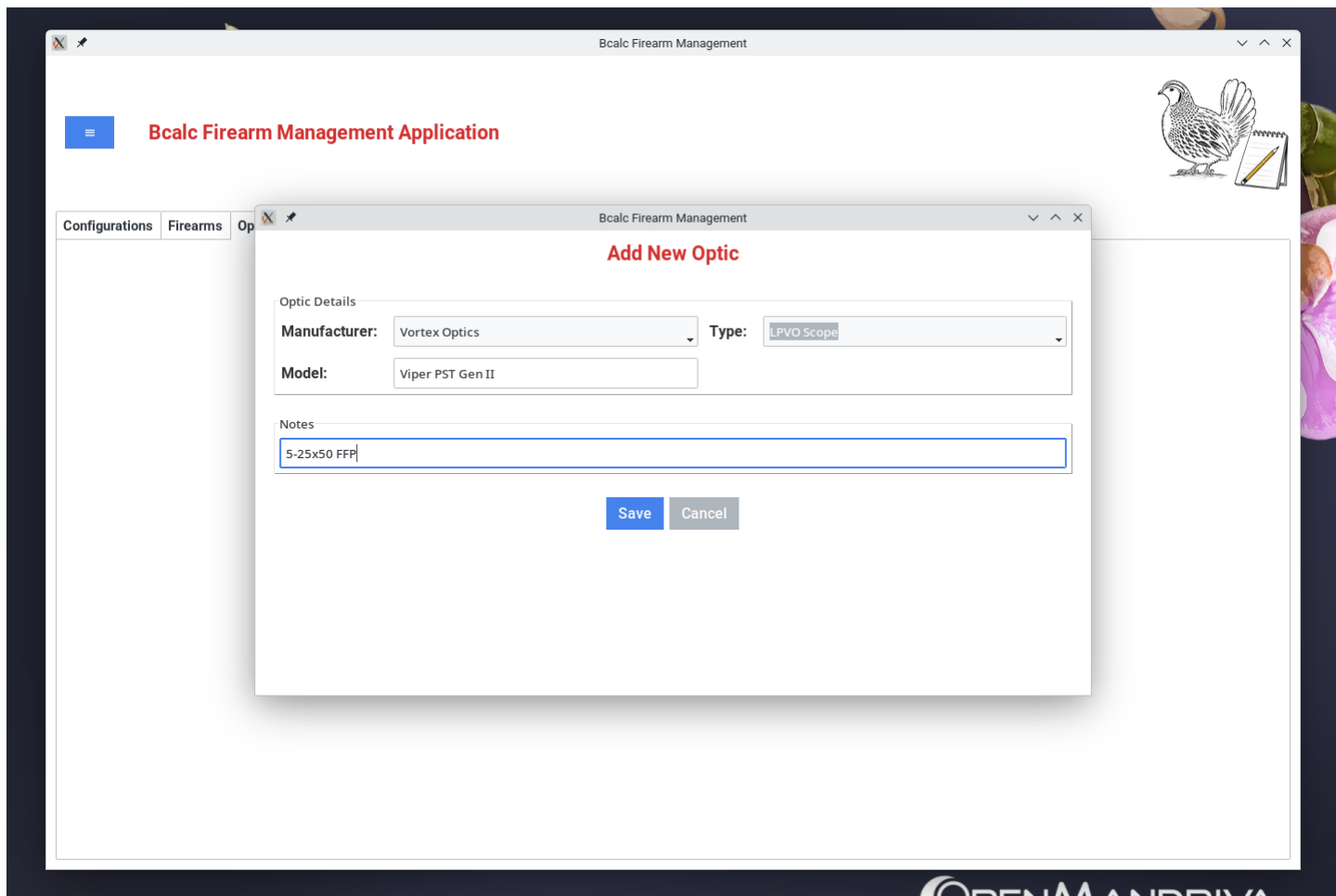
Photo & Notes

Firearm Photo:

Notes:

Step 2. Optics

Next, you can enter an optic. Select the Optic Tab and then the Add Optics button.



The screenshot displays the Bcalc Firearm Management Application interface. The main window has a title bar "Bcalc Firearm Management" and a header area with a menu icon and the text "Bcalc Firearm Management Application". A sidebar on the left contains tabs for "Configurations", "Firearms", and "Optics". The "Optics" tab is selected. A modal dialog box titled "Add New Optic" is open in the center. It contains the following fields:

- Optic Details**
 - Manufacturer:** A dropdown menu with "Vortex Optics" selected.
 - Type:** A dropdown menu with "LPVO Scope" selected.
 - Model:** A text input field containing "Viper PST Gen II".
- Notes**
 - A text input field containing "5-25x50 FFP".

At the bottom of the dialog box are two buttons: "Save" (blue) and "Cancel" (grey). The background of the application window shows a faint illustration of a turkey and some flowers.

Step 3. Ammunition

You guessed it. Enter ammunition by selecting the Ammo Tab and then the Add Ammo button. Both purchased and hand-load ammunition can be tracked.

Bcalc Firearm Management

Add New Ammo Record

Basic Information

Manufacturer: Caliber/ Gauge:

Grains (Proj.): Projectile Type:

Shot Size: # of Pellets:

Ammo Box Name: Use Case:

Drag Function: Ballistic Coeff.:

☐ Do Not Rebuy (Poor Performance)

Reloading Components (Optional)

Bullet Mfg: Powder Mfg:

Bullet Name: Powder Name:

Brass Mfg: Powder Lot:

Case Weight: Powder Grains:

Overall Length: Primer Mfg:

CBTO: Primer Type:

Crimp Type: Primer Lot:

Hull Mfg: Wad Mfg:

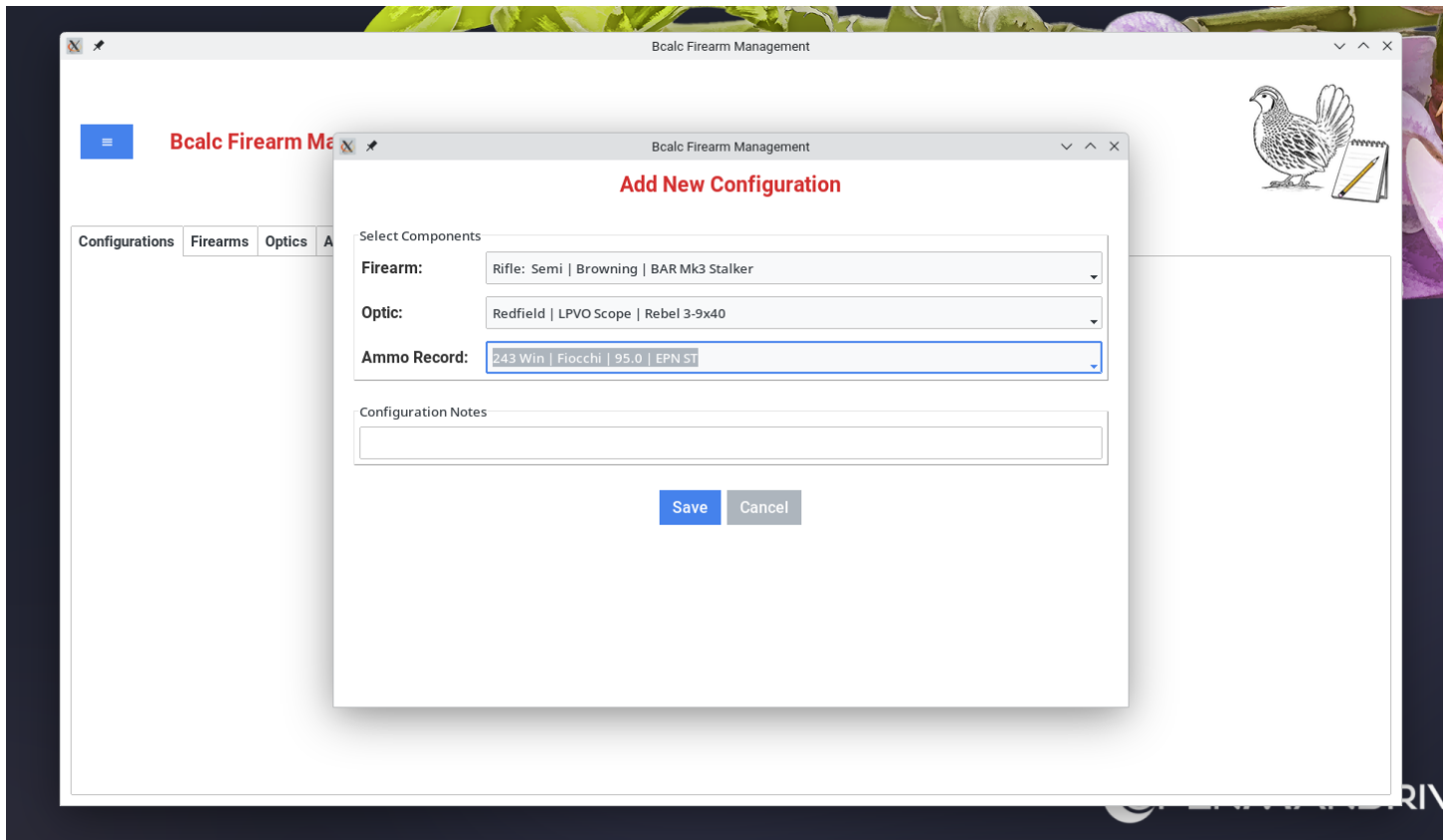
Batch Stamps

Batch Quantity: Date Loaded (YYYY-MM-DD):

Notes

Step 4. Configurations

Enter a configuration by selecting the Configuration Tab and then the Add Configuration button. At a minimum, a configuration must include a firearm. There are no practical limits to the number of different configurations of firearms, optics, and ammunition. It is very likely that each firearm will have several ammo brands associated with it.



The screenshot displays the Bcalc Firearm Management application interface. A modal dialog box titled "Add New Configuration" is open, allowing users to create a new setup. The dialog includes three dropdown menus under the "Select Components" section: "Firearm:" (set to "Rifle: Semi | Browning | BAR Mk3 Stalker"), "Optic:" (set to "Redfield | LPVO Scope | Rebel 3-9x40"), and "Ammo Record:" (set to "243 Win | Fiocchi | 95.0 | EPN ST"). Below these is a text area for "Configuration Notes" and two buttons, "Save" and "Cancel". The background application window shows a sidebar with tabs for "Configurations", "Firearms", "Optics", and "Ammunition", and a header with the application name and a quail icon.

Step 5. Range Log

Select the Range Log Tab and the add Session button. Configurations are the primary key for the range results. All other data is optional. A Rating field is provided to help track the firearm-ammo performance.

Bcalc Firearm Management

Add New Range Session

Session Details

Configuration:

Date (YYYY-MM-DD): Location:

Performance & Condition Data

Rounds Fired:	80	Target Range:	100	Wind Speed:	
Avg Muzzle Vel:	3240	Avg MOA:	1.4	Wind Dir (0-359):	
Max Muzzle Vel:	3260	Best MOA:	1.0	Temperature:	
Std Dev Vel:		Std Dev MOA:		Humidity:	
Ext Spread Vel:		Ext Spread MOA:		Rating:	unknown

Photo & Notes

Target Photo:

Notes:

Step 6. Reports

Select the Reports Tab. Several data points are shown including total rounds per firearm and the top configurations per caliber calculated using MOA and Std Dev MOA data.

Bcalc Firearm Management



Bcalc Firearm Management Application

ConfigurationsFirearmsOpticsAmmoRange LogReports

Firearm Round Stats

Select Firearm: Browning BAR Mk3 Stalker

Total Rounds Fired: 195

Top Ammo: Fiocchi EPN ST (243 Win)

Top Configurations by Caliber

Select Caliber: 243 Win

Rank	Configuration	Rating	Rounds
1	Browning BAR Mk3 Stalker Redfield RebNo Rating		195
2	Ruger American Gen II Predator VortexNo Rating		85

Configuration Stats

Select Configuration: Henry H9 Provider | Leupold | Hornady

Avg Velocity: 2465 fps

Avg MOA: 1.50

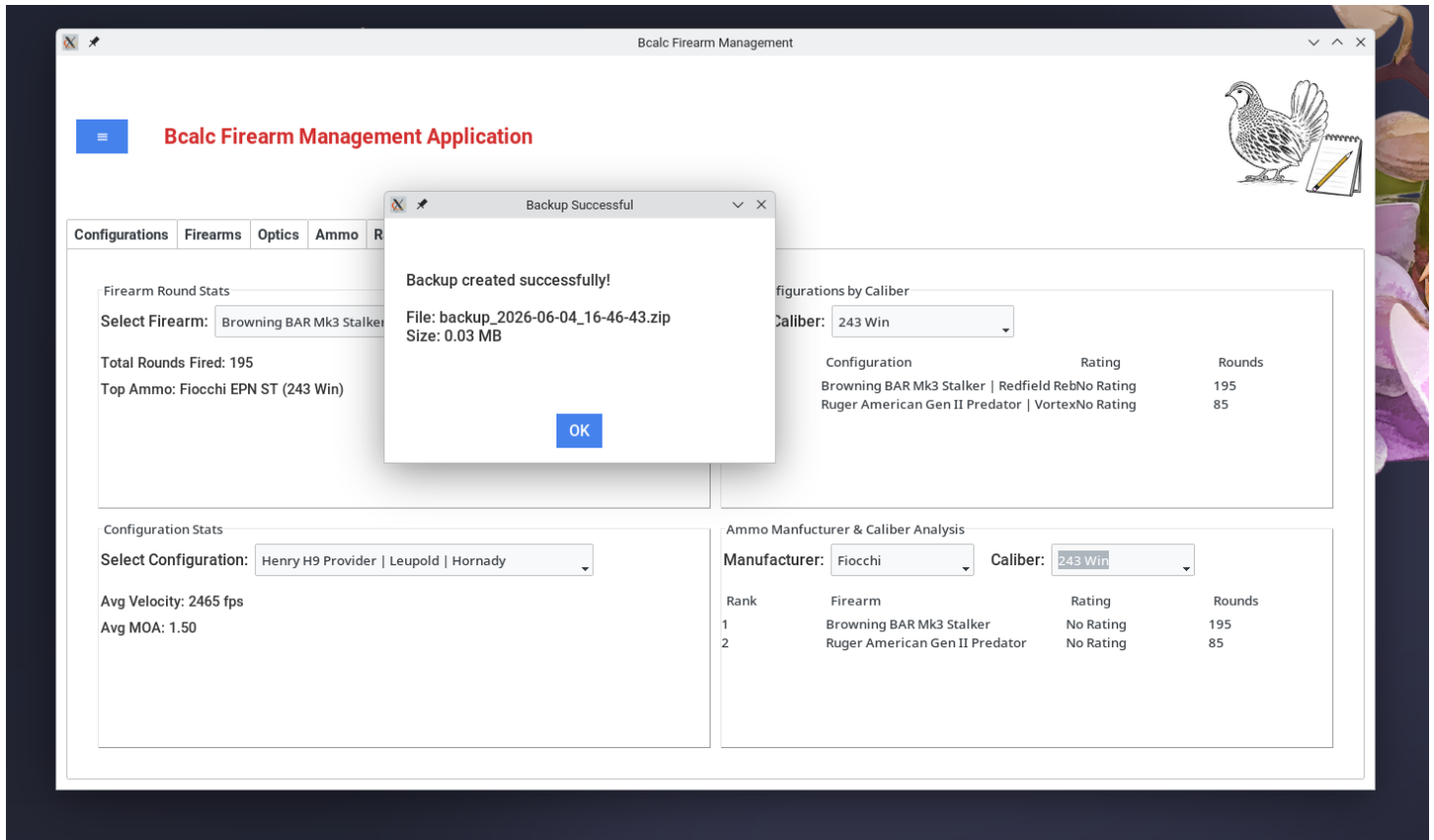
Ammo Manufacturer & Caliber Analysis

Manufacturer: FiocchiCaliber: 243 Win

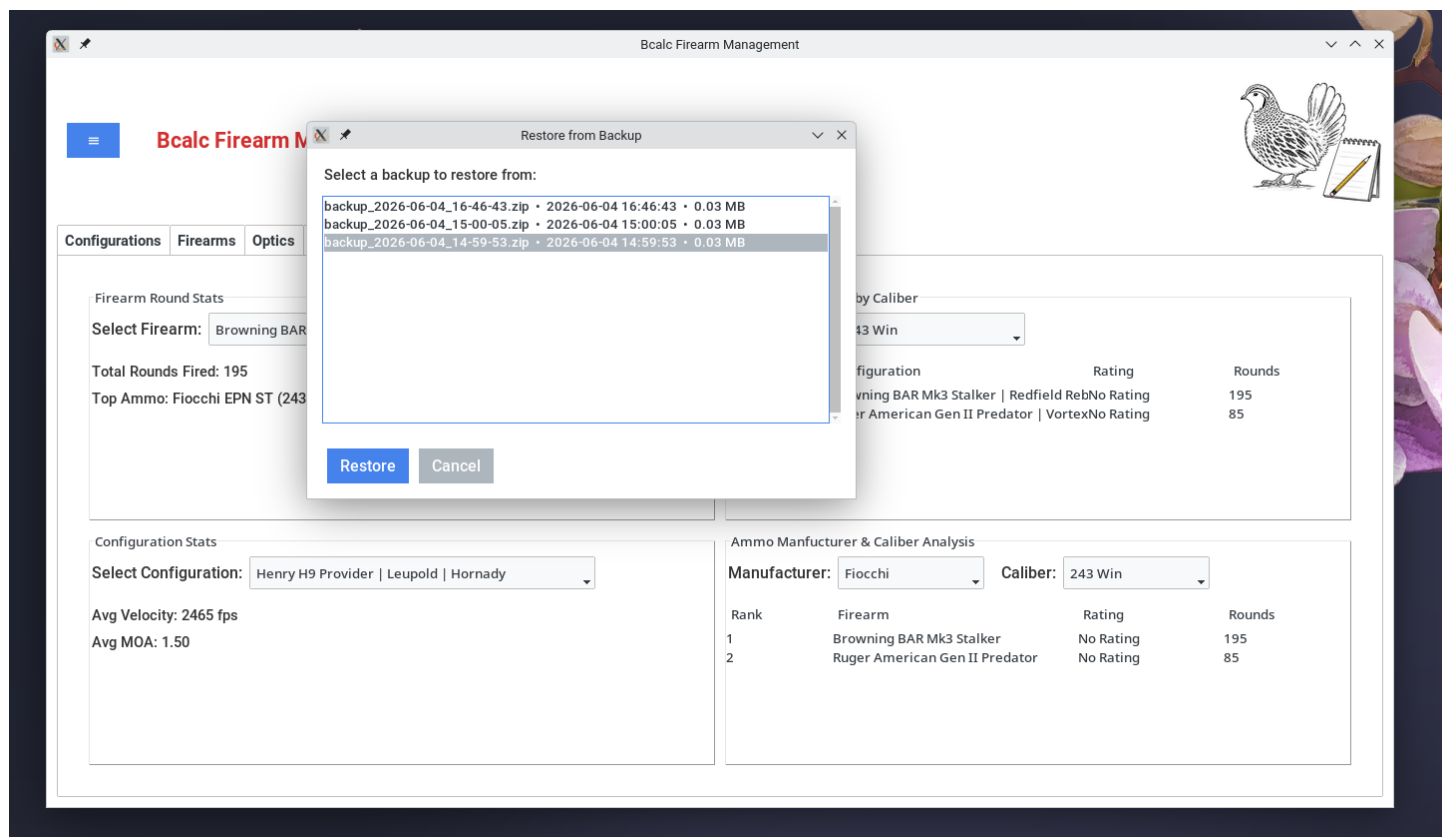
Rank	Firearm	Rating	Rounds
1	Browning BAR Mk3 Stalker	No Rating	195
2	Ruger American Gen II Predator	No Rating	85

Step 7. Optional: Backup and Restore

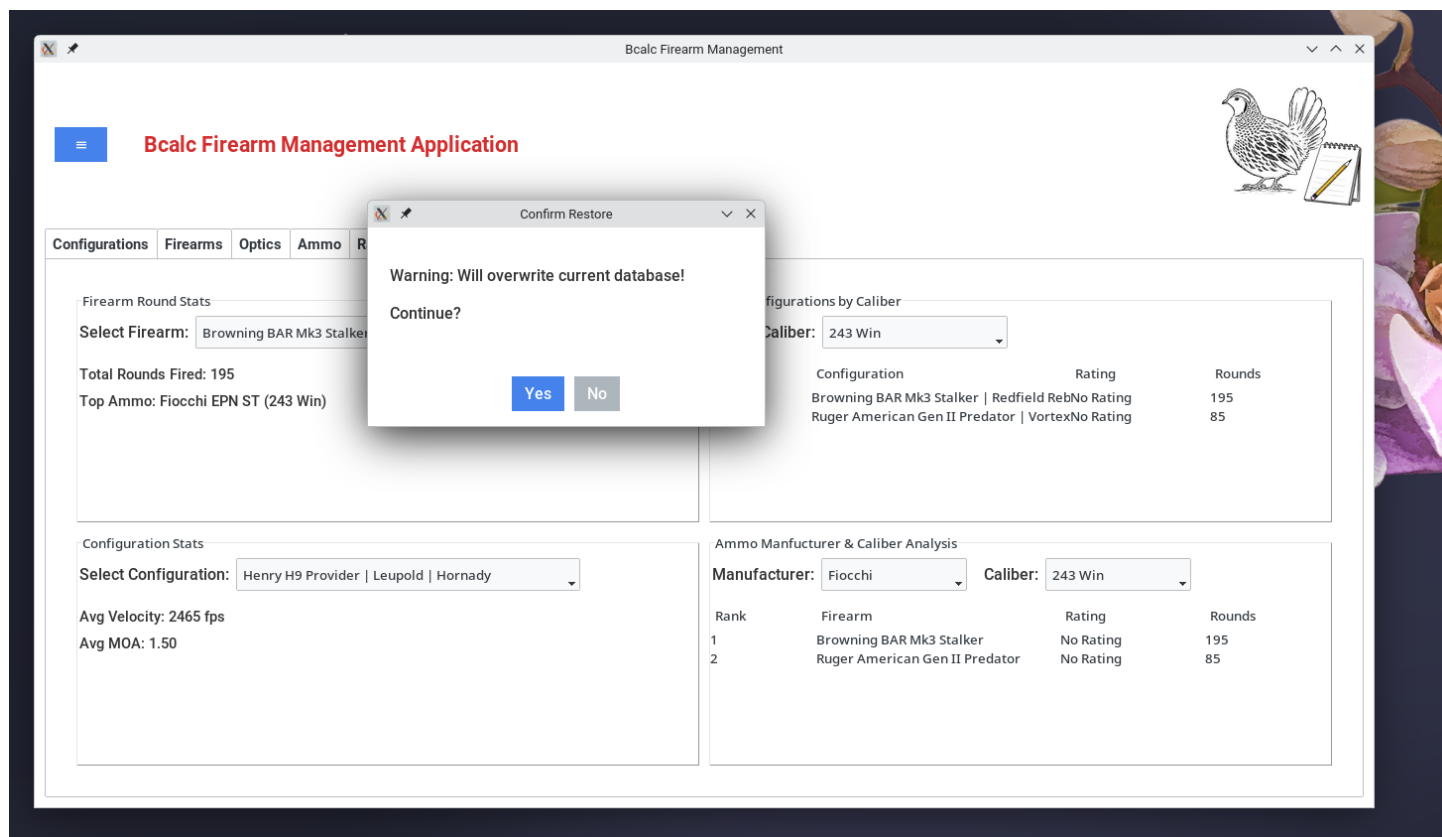
It's a good idea to periodically save a copy of the database just in case. Click the blue hamburger menu at the top left of the window and select Backup Database.



To restore your records from a backup, click the blue hamburger menu and select Restore Database. Pick a backup and press Restore.



Your current database will be replaced by the backup so be certain you want to restore your previous records.



5. Credits & License

The Bcalc Firearm Management App was created by Brian Calc.

GNU General Public License (GPL) Version 3.